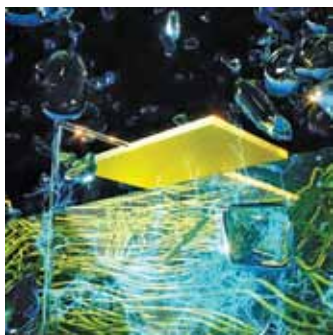




autonomous off-road driving technology. This innovative addition aims to enhance off-roading experiences for seasoned enthusiasts and newcomers. It also assists Jeep sport utility vehicle (SUV) owners navigate challenging everyday driving conditions. The Jeep brand is actively leading the way in the development of automated off-road driving technology while also solidifying its position as a leader in off-road electrification through the expansion of its one of the SUV lineups.

Continuous electricity generation from air humidity in any material made possible

Researchers at the University of Massachusetts Amherst have discovered a breakthrough method which enables continuous



Nanopores are the secret to making electricity from thin air. These nanopores allow water molecules to pass through and create a charge imbalance, essentially forming a battery that runs as long as there is humidity (Credit: Derek Lovley/Ella Maru Studio)

electricity generation from air humidity, utilising any material featuring nanopores smaller than 100 nanometres. The researchers have created a small-scale cloud that reliably captures and harvests electricity from the air using the “generic Air-gen effect.”

It builds on the discovery of harnessing electricity from the air using protein nanowires. The Air-gen effect is universal and applicable to any material with sub-100nm pores,

corresponding to the distance water molecules travel before colliding. The team has created an electricity harvester using sub-100nm nanopores. Like clouds, water molecules colliding with pore edges generate a charge imbalance. This enables a humidity-powered battery with an indefinite operation.

Breakthrough made in ML models for speech recognition and object detection

MIT and IBM Watson Artificial Learning Lab researchers have developed a new method to analyse unlabelled audio and visual data, improving machine learning (ML) models for speech recognition and object detection. The researchers evaluated contrastive audio-visual masked autoencoder (CAV-MAE), their contrastive loss and masked autoencoder-free method, and other techniques on standard datasets. Tasks included audio-visual retrieval and event classification. Retrieval involved finding missing components while classification identified actions or sounds. CAV-MAE outperformed prior techniques by 2% for event classifica-

tion, matching industry-level models. It ranked similarly to models using only contrastive loss. Incorporating multi-modal data in CAV-MAE improved single-modality representation and audio-only event classification. Researchers see CAV-MAE as a breakthrough for applications adopting multi-modality and audio-visual fusion.

Solar, wireless sensor developed to monitor bridge deformation continuously

Researchers at Drexel University’s College of Engineering have developed a solar-powered, wireless sensor system



Wireless, solar-powered sensors developed by Drexel engineers could allow for continuous monitoring of bridges (Credit: Drexel University)

to monitor bridge deformation

continuously and notify authorities of significant performance deterioration. The wireless displacement sensor comprises a solar cell, a displacement poten-

tiometer, and a monitoring transceiver. It measures bridge deformation during traffic, transmitting data to a remote station. The lightweight potentiometer detects girder movement, identifying structural issues. Powered by solar cells and backup batteries, multiple sensors can be wirelessly installed on the bridge, supporting various monitoring sensors for acceleration, tilt, and displacement. The system’s main advantage is cable elimination, reducing costs and installation efforts. The wireless platform enables readings from various sensor types simultaneously. The team crafted a weather-resistant power supply featuring a 10W solar cell and a lithium-ion battery and tested it in harsh winter conditions to ensure enduring performance.

A megawatt motor being designed to electrify large aircrafts

MIT engineers are developing a 1-megawatt motor, a crucial milestone for electrifying larger aircraft. The team has designed, tested, and computed the motor’s components, confirming its ability to generate one megawatt of power. The motor comprises a high-speed rotor with magnet arrays, a compact low-loss stator with copper windings, an efficient heat exchanger for cooling and torque transmission, and a distributed power electronics system with custom circuit boards for precise current control. The motor system integrates circuit boards closely with the machine, reducing loss and enabling efficient air cooling via an integrated heat exchanger. The high-frequency switch-